

I am an undergraduate student majoring in Bioinformatics at Tongji University, with broad and sustained interests in artificial intelligence, computational biology, and generative modeling. At the intersection of these fields, I hope to explore how mathematical and computational methods can deepen our understanding of living systems. I aim to translate that understanding into scientific discoveries that can be tested and put into practice.

My research interests include virtual cell modeling, molecular generation for drug discovery, and the development of AI agents for biological research. Methodologically, I am particularly interested in diffusion models and flow matching. I am drawn to their elegant mathematical formulations and their capacity to model complex distributions, construct continuous generative processes, and explore high-dimensional spaces. Much of my current work centers on these generative approaches. Looking ahead, I hope to improve their modeling capabilities and computational efficiency, developing theoretically grounded methods that address practical problems in biology.

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Google Scholar: <https://scholar.google.com/citations?user=crCPHUYAAAAJ&hl=zh-CN>

GitHub: <https://github.com/XuQisheng062>